SOFT L-TOPOLOGICAL LANGUAGE MODELLING

A Thesis

Submitted in partial fulfilment for the Degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

by

RITHESH S S (CB.AI.U4AID23031)

ASHWIN J R (CB.AI.U4AID23029)

VISHWA VINAYAK C K (CB.AI.U4AID23051)

PRANAV SIVAKUMAR (CB.AI.U4AID23028)

Team AA03



DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS (INDIA)

October 2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS



BONAFIDE CERTIFICATE

This is to certify that the thesis entitled "**Soft L-Topological Language Modelling**" submitted by **Rithesh S S** (CB.AI.U4AID23031), **Ashwin J R** (CB.AI.U4AID23029), **Vishwa Vinayak C K** (CB.AI.U4AID23051), and **Pranav Sivakumar** (CB.AI.U4AID23028) for the award of the **Degree of Bachelor of Technology in Artificial Intelligence and Data Science** under the **Faculty of Engineering** is a bonafide record of the original research work carried out under my guidance and supervision at the "Department of Artificial Intelligence", Amrita Vishwa Vidyapeetham, Coimbatore Campus.

Submitted for the examination held on October 23, 2026.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Dr. Sandhya S. Pai
Thesis Advisor / Assistant Professor,
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE CAMPUS

DECLARATION

We, **Rithesh S S** (CB.AI.U4AID23031), **Ashwin J R** (CB.AI.U4AID23029), **Vishwa Vinayak C K** (CB.AI.U4AID23051), and **Pranav Sivakumar** (CB.AI.U4AID23028), hereby declare that this thesis entitled "**Soft L-Topological Language Modelling**" is the record of original work conducted by us under the guidance of **Dr. Sandhya S. Pai**, Assistant Professor, Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus. To the best of our knowledge, this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or similar title in this or any other University.

Place: Coimbatore **Signature of the Students**

Date: October 23, 2026

COUNTERSIGNED

Dr. K. P. Soman
Professor and Dean,
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus,
India.

Contents

Acknowledgements **ii**

List of Figures **iii**

List of Tables **iv**

Abbreviations and Symbols **v**

Abstract **vi**

1 Introduction **1**
1.1 Background and Motivation 1
1.2 Core Idea: Multi-Attribute Continuous Logic 1
1.3 Problem Statement and Objectives 2
1.4 Summary of Key Findings and Outline 2

2 Theoretical Foundations and Related Work **3**
2.1 The Standard Transformer and Softmax Bottleneck 3
2.2 Prior Approaches to Sparsity 3
2.3 Soft Set Theory and Continuous Lattices 4
2.4 Triangular Norms and the Nilpotent Advantage 4

3 Proposed Soft L-Topological Architecture **5**
3.1 System Architecture Overview 5
3.2 Bounded Continuous Lattice Embeddings 5
3.3 The Continuous Lattice Sequence Router 6
3.4 Continuous Lattice KV-Caching 7

4 Experimental Setup and Foundational Ablations **8**
4.1 Experimental Methodology and Controlled Parity 8
4.2 Foundational Empirical Ablations (Experiments 1 to 5) 9

5 Large-Scale Benchmarks and Critical Analysis **10**
5.1 Large-Scale Benchmarks (Experiments 6 to 10) 10
5.2 Critical Discussion and Explainability 12

6 Conclusion and Future Scope **13**
6.1 Summary of Contributions 13
6.2 Practical Implications and Future Roadmap 13

A Mathematical Proofs and Specifications **14**
A.1 Mathematical Proofs of Nilpotent Triangular Norm Properties 14
A.2 Automated Unit Testing Suite (19 Passing Tests) 15
A.3 Reproducibility Protocol and Environment Setup 15

References **16**

Acknowledgements

We express our profound gratitude to our project supervisor and mentor, **Dr. Sandhya S. Pai**, Assistant Professor in the Department of Artificial Intelligence, whose pioneering mathematical research on soft lattice topological spaces (Pai & Baiju, 2021) provided the theoretical foundation for this thesis. Her continuous intellectual guidance, mathematical insights, and constructive reviews were invaluable throughout the conception, implementation, and benchmarking of this architecture.

We extend our sincere thanks to **Dr. K. P. Soman**, Professor and Dean, Department of Artificial Intelligence. We also thank our peers, faculty members, and families for their constant encouragement.

RITHESH S S
ASHWIN J R
VISHWA VINAYAK C K
PRANAV SIVAKUMAR
(Team AA03)

List of Figures

4.1 Exp 2: Operator Pareto Frontier. 9
4.2 Exp 4: Threshold τ vs. Sparsity. 9

5.1 Exp 7: Distractor Scaling Curve. 11
5.2 Exp 8: Deductive Transitivity. 11
5.3 Exp 9: High-Volume Scaling. 11
5.4 Exp 10: Encoder MLM Benchmarks. 11

List of Tables

2.1 Canonical Continuous Triangular Norm Families 4

4.1 Architectural Specifications and Parameter Parity Comparison 8

5.1 Experiment 7: Distractor Scaling Needle-in-a-Haystack Results 10
5.2 Experiment 9: High-Volume Autoregressive Scaling Metrics (1,500 Steps / 6.14M Tokens) 11

A.1 Comprehensive Unit Testing Suite (19 Passing Automated Tests) 15

Abbreviations and Symbols

Abbreviations

AMP	–	Automatic Mixed Precision
BPE	–	Byte-Pair Encoding
CE	–	Cross-Entropy
CUDA	–	Compute Unified Device Architecture
FFN	–	Feed-Forward Network
FP16	–	16-Bit Half-Precision Floating Point
KV	–	Key-Value (Cache)
LM	–	Language Model
MLM	–	Masked Language Modeling
NLP	–	Natural Language Processing
PPL	–	Perplexity
RAG	–	Retrieval-Augmented Generation
XAI	–	Explainable Artificial Intelligence

Mathematical Symbols

$[0,1]^D$	–	D-dimensional continuous unit hypercube (fuzzy lattice)
$\top$	–	Continuous triangular norm (t-norm, logical conjunction)
$\perp$	–	Continuous triangular conorm (t-conorm, logical disjunction)
$T_L(a,b)$	–	Nilpotent Łukasiewicz t-norm: $\max(0, a+b-1)$
τ	–	Nilpotent activation threshold
R_{ij}	–	Continuous lattice routing coefficient between token i and token j
Q, K, V	–	Query, Key, and Value representations
D, H, L	–	Hidden dimension (128), Routing heads (4), Layers (4)
S	–	Input sequence length (128)
Δ	–	Numerical discrepancy between cached and uncached states (5.06×10^{-7})
ϵ	–	Numerical stabilizer constant (10^{-7})

Abstract

Standard autoregressive language models rely fundamentally on dot-product attention followed by the softmax operator. While empirically successful, the mathematical structure of softmax enforces a strictly positive probability mass across all context tokens, creating distractor noise leakage and precluding true mathematical sparsity.

In this thesis, we propose and investigate the **Soft L-Topological Language Model**, an alternative neural architecture that replaces statistical Euclidean vectors and softmax routing with continuous soft set representations bounded strictly in $[0, 1]^D$ and differentiable nilpotent lattice operators derived from the Łukasiewicz t-norm ($T_L(a, b) = \max(0, a + b - 1)$). When multi-attribute overlap falls below a threshold τ, connections drop to **identically 0.0000**. Across an extensive 10-experiment empirical benchmark spanning 6.14 million tokens across WikiText-2 and TinyStories, we present an unembellished assessment:

- **Perplexity Trade-off:** Soft L-Topology converges reliably (validation loss 3.7455 / PPL 42.33 on TinyStories), but does not outperform the standard baseline (loss 3.4866 / PPL 32.67), as softmax provides a smoother surface for cross-entropy log-loss.
- **Deterministic Sparsity:** In return, Soft L-Topology achieves a sustained **55.06% exact-zero routing sparsity** ($R_{ij} \equiv 0.0000$) throughout generation.
- **Distractor Immunity:** Under 16-distractor Needle-in-a-Haystack tests, Soft L-Topology prunes 66.9% of distractor connections to exact zero, reducing spurious attention leakage by **49.4% in absolute terms**.
- **Logical Entailment & Encoder Edge:** Soft L-Topology doubles Modus Ponens deduction accuracy (40.0% vs. 20.0%) and achieves lower Masked Language Modeling loss (6.2041 vs. 6.2422) with 50.0% distractor pruning.

Chapter 1

Introduction

1.1 Background and Motivation

Modern Natural Language Processing (NLP) is dominated by autoregressive language models built on the Transformer architecture [1]. Transformers achieve scalability and expressive power through self-attention, which maps input sequences to contextualized representations via scaled dot products normalized by the *softmax* operator:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad \text{where softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{1.1}$$

While empirically successful, softmax enforces a rigid mathematical constraint: because $\exp(z) > 0$ for all finite real numbers, **softmax assigns a strictly positive probability to every token in the sequence**. Consequently, standard self-attention enforces a 100% dense connectivity graph. Every token—regardless of whether it is an essential grammatical argument or an irrelevant, noisy distractor—receives non-zero attention mass. This produces two structural vulnerabilities:

1. **Distractor Noise Leakage:** In long-context reasoning, documents inevitably contain irrelevant paragraphs, distractor facts, and conversational filler. Because softmax cannot output a true zero (0.0000), distractor tokens continuously siphon attention mass away from the core context, compounding across layers and increasing factual hallucinations.

2. **Absence of Faithful Sparsity:** Pairwise attention over a context of length S requires dense $O(S^2)$ memory and computation. Even when 90% of tokens share no semantic relation to the current query, the model cannot prune connections to absolute zero.

1.2 Core Idea: Multi-Attribute Continuous Logic

To address this limitation, we rethink how tokens interact. Instead of asking *"Are two unconstrained vector arrows pointing in similar directions in Euclidean space $\mathbb{R}^D$?"*, we ask: *"To what degree does token A satisfy the semantic attributes required by token B in a bounded space $[0, 1]^D$?"*

In multi-attribute continuous logic, relationships are governed by logical containment (intersection). When an entity lacks the attributes required to satisfy a relation, the conjunction reaches a **hard logical floor at zero**. To realize this floor differentiably, we employ the **nilpotent Łukasiewicz triangular norm (t-norm)** [2]:

$$T_L(a, b) = \max(0, a + b - 1) \tag{1.2}$$

If the sum of two attribute memberships $(a + b)$ falls below 1, the result evaluates to **identically 0.0000**. By introducing an activation threshold τ, this operator acts as a continuous, differentiable

gate: when semantic alignment is insufficient, the connection is mathematically severed.

1.3 Problem Statement and Objectives

This thesis investigates the following core research question:

> *Can continuous soft set representations in $[0,1]^D$ and nilpotent triangular norms produce a differentiable sequence router that achieves deterministic exact-zero sparsity and distractor immunity without sacrificing linguistic fluency?*

To answer this, we pursue four concrete objectives: (1) Formulate a continuous lattice router using bounded convex embeddings, nilpotent Łukasiewicz t-norms, and Jaccard containment; (2) Benchmark a parameter-matched 13.67M prototype on WikiText-2 and TinyStories under strict parity with a Transformer; (3) Quantify resilience against context distractors (Needle-in-a-Haystack from 1 to 16 distractors) and multi-step deduction (Modus Ponens); and (4) Honestly analyze the trade-off between exact-zero sparsity and next-token perplexity.

1.4 Summary of Key Findings and Outline

Adhering strictly to empirical honesty, our findings demonstrate: (1) *Perplexity Trade-Off:* Soft L-Topology does not beat standard Transformers in raw perplexity (TinyStories validation loss 3.7455 / PPL 42.33 vs. Baseline 3.4866 / PPL 32.67), as softmax provides a smoother surface for cross-entropy log-loss; (2) *Deterministic Sparsity:* Soft L-Topology sustains **55.06% exact-zero routing sparsity** throughout autoregressive generation; (3) *Distractor Immunity:* Under 16-distractor injection, Soft L-Topology prunes 66.9% of distractor connections to exact zero, reducing spurious leakage by **49.4% in absolute terms**; and (4) *Deductive Entailment & Encoder Edge:* It doubles Modus Ponens deduction accuracy (40.0% vs. 20.0%) and achieves lower validation loss in bidirectional Masked Language Modeling (6.2041 vs. 6.2422) with 50.0% distractor pruning.

The remainder of this thesis is organized as follows: Chapter 2 reviews theoretical foundations and related work; Chapter 3 details the Soft L-Topological architecture and lattice KV-caching; Chapter 4 presents experimental setup and foundational ablations (Exps 1–5); Chapter 5 details large-scale benchmarks (Exps 6–10) and critical discussion; Chapter 6 concludes with practical implications; and Appendix A provides mathematical proofs and technical specifications.

Chapter 2

Theoretical Foundations and Related Work

2.1 The Standard Transformer and Softmax Bottleneck

The canonical Transformer architecture [1] maps an input token sequence $(w_1, \ldots, w_S)$ to contextualized representations via multi-head self-attention. For sequence representations $X \in \mathbb{R}^{S \times D}$, linear projection matrices produce Queries (Q), Keys (K), and Values (V): $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. Pairwise compatibility is computed via scaled dot products normalized row-wise by softmax:

$$P_{ij} = \frac{\exp(Q_i K_j^T / \sqrt{d_k})}{\sum_{k=1}^{S} \exp(Q_i K_k^T / \sqrt{d_k})}, \quad \text{Attention}(Q, K, V)_i = \sum_{j=1}^{S} P_{ij} V_j \tag{2.1}$$

Proposition 2.1 (Strict Positivity of Softmax)**.** Let $\mathbf{z} \in \mathbb{R}^S$ be any finite vector of logits. Then for all $j \in \{1, \ldots, S\}$:

$$P_j = \frac{\exp(z_j)}{\sum_{k=1}^{S} \exp(z_k)} > 0 \tag{2.2}$$

Output probability $P_j = 0$ if and only if $z_j \to -\infty$.

In digital computing with finite floating-point numbers, logits are always finite; hence, $P_{ij} > 0$ universally. In long contexts containing hundreds of tokens, completely irrelevant tokens continuously receive fractional weights (10^{-3} to 10^{-6}). Over L sequential layers, this infinitesimal leakage compounds, blending distractor features into factual representations. As surveyed by Ji et al. [3], this dense contextual leakage is a primary architectural driver of factual hallucination and unfaithful polysemy resolution in standard autoregressive models.

2.2 Prior Approaches to Sparsity

Researchers have previously explored two main strategies to introduce attention sparsity:

- **Heuristic Structural Sparsity:** Longformer [4] and BigBird [5] restrict attention to sliding local windows and dilated strides. While reducing complexity, these patterns are static and predefined; they cannot dynamically form or sever connections based on semantic content.
- **Mathematical Activation Sparsity:** Sparsemax [6] computes Euclidean projections onto the simplex to induce zeros. However, when coordinates fall below the threshold, their gradients become identically zero, causing optimization instability and "dead neuron" regions. Entmax [7] interpolates smoothly via Tsallis entropy, but requires iterative numerical solvers that introduce latency on parallel GPU hardware.

2.3 Soft Set Theory and Continuous Lattices

To establish a principled, content-dependent routing mechanism, we turn to **Soft Set Theory**, introduced by Molodtsov [8] as a mathematical framework for parameterized uncertainty.

Definition 2.1 (Soft Set)**.** Let U be a universe of objects and E a set of parameters. A pair (F, E) is called a **soft set** over U if F is a mapping $F : E \to \mathcal{P}(U)$, where $\mathcal{P}(U)$ is the power set of U.

Maji et al. [9] extended this formulation to *Fuzzy Soft Sets*, where $F(e)$ assigns membership values in $[0, 1]$. Building upon these foundations, Pai and Baiju [10] formulated *Soft Lattice Topological Spaces*, establishing a unified mathematical framework connecting parameterized soft sets to continuous lattice structures. In this thesis, we adapt soft lattice topological principles to neural sequence modeling: every discrete token w is mapped to a continuous soft set over a bounded lattice hypercube: $\mathbf{x} \in [0, 1]^D$, where each coordinate $x_d \in [0, 1]$ represents the truth value or satisfaction degree of a latent semantic attribute.

2.4 Triangular Norms and the Nilpotent Advantage

To compute logical conjunctions (intersections) on $[0, 1]^D$, we employ **Triangular Norms (t-norms)** [2], which satisfy commutativity, associativity, monotonicity, and boundary conditions ($T(a, 1) = a, T(a, 0) = 0$).

Table 2.1: Canonical Continuous Triangular Norm Families

Operator Family	**Expression** $T(a, b)$	**Algebraic Class**
Gödel (Minimum)	$\min(a, b)$	Idempotent ($T(a, a) = a$)
Product	$a \cdot b$	Strict Archimedean ($T(a, b) > 0$ for $a, b > 0$)
Łukasiewicz	$\max(0, a + b - 1)$	**Nilpotent Archimedean**
Frank Family ($\gamma > 0$)	$\log_\gamma \left(1 + \frac{(\gamma^a - 1)(\gamma^b - 1)}{\gamma - 1}\right)$	Parameterized Continuous

A t-norm is **nilpotent** if there exist non-zero elements $a, b > 0$ such that $T(a, b) = 0$. For Łukasiewicz, $T_L(a, b) = \max(0, a + b - 1) = 0 \iff a + b \leq 1$. Unlike Gödel or Product, which remain strictly positive whenever inputs are positive, the Łukasiewicz operator provides a **natural, continuous mathematical floor at zero**. Crucially, when $a + b > 1$, its gradient is $\frac{\partial T_L}{\partial a} = 1.0$, $\frac{\partial T_L}{\partial b} = 1.0$, providing a **unit gradient** across its active domain, combining un-saturating backpropagation with deterministic exact-zero sparsity.

Chapter 3

Proposed Soft L-Topological Architecture

3.1 System Architecture Overview

The Soft L-Topological Neural Network is an autoregressive and bidirectional language modeling architecture that replaces unconstrained Euclidean vector spaces and dense softmax routing with **bounded continuous soft set representations** and **nilpotent lattice sequence routing**.

The overall forward data flow proceeds as follows:

1. Input discrete tokens $w_t \in \mathcal{V}$ are embedded directly into the continuous unit hypercube $[0,1]^D$.
2. Bounded learnable positional encodings in $[0,1]^D$ are combined with token representations via convex blending.
3. A stack of L identical **Lattice Blocks** processes the representations, each combining a multi-head continuous lattice router, residual connections, and position-wise feed-forward networks (FFN).
4. A final linear projection computes next-token logits over the vocabulary.

3.2 Bounded Continuous Lattice Embeddings

To apply continuous logic, representations must remain strictly bounded in $[0,1]^D$ without arbitrary magnitude drift.

3.2.1 Convex Token and Positional Blending

Raw embedding parameters $\widetilde{E} \in \mathbb{R}^{|\mathcal{V}| \times D}$ and positional parameters $\widetilde{P} \in \mathbb{R}^{S \times D}$ are normalized via element-wise sigmoid:

$$\mathbf{e}_{\text{tok}}(w) = \sigma(\widetilde{E}_w) \in (0,1)^D, \quad \mathbf{e}_{\text{pos}}(t) = \sigma(\widetilde{P}_t) \in (0,1)^D \tag{3.1}$$

Instead of unconstrained addition ($x_{\text{tok}} + x_{\text{pos}}$), they are combined via **convex blending** with coefficient $\lambda = 0.5$:

$$\mathbf{x}_t^{(0)} = \lambda \mathbf{e}_{\text{tok}}(w_t) + (1-\lambda)\mathbf{e}_{\text{pos}}(t) \tag{3.2}$$

Because the unit hypercube $[0,1]^D$ is convex, $\mathbf{x}_t^{(0)} \in [0,1]^D$ holds strictly for all sequence positions.

3.2.2 Beta Distribution Initialization

Standard Gaussian initialization centered at zero produces uniform memberships near 0.5 under sigmoid. To encourage diverse attribute specialization, raw embedding weights are initialized using an asymmetric **Beta distribution**:

$$\widetilde{E}_{vd} \sim \text{logit}(\text{Beta}(\alpha, \beta)), \quad \alpha = 1.0,\ \beta = 3.0 \tag{3.3}$$

This concentrates initial memberships between 0.1 and 0.4, setting most attribute pairs near or below the nilpotent threshold at initialization.

3.3 The Continuous Lattice Sequence Router

Given sequence representations $X \in \mathbb{R}^{S \times D}$, the router projects hidden states into H parallel heads ($d_k = D/H$):

$$Q^{(h)} = \sigma(XW_Q^{(h)}) \in [0,1]^{S \times d_k} \qquad \text{(Queries as fuzzy attribute requirements)} \tag{3.4}$$

$$K^{(h)} = \sigma(XW_K^{(h)}) \in [0,1]^{S \times d_k} \qquad \text{(Keys as fuzzy attribute satisfactions)} \tag{3.5}$$

$$V^{(h)} = XW_V^{(h)} \in \mathbb{R}^{S \times d_v} \qquad \text{(Values as unconstrained semantic features)} \tag{3.6}$$

Notice that Queries and Keys are bounded in $[0,1]^{S \times d_k}$ to serve as truth membership functions, while Values ($V^{(h)}$) remain in unconstrained Euclidean space $\mathbb{R}^{S \times d_v}$ to preserve rich feature capacity.

3.3.1 Thresholded Łukasiewicz Conjunction

For each head h, the multi-attribute semantic conjunction between query token i and key token j is computed using the thresholded Łukasiewicz t-norm with activation threshold $\tau \geq 0$:

$$\mathcal{M}_{ijd}^{(h)} = \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau) \tag{3.7}$$

Total attribute overlap is the sum of active conjunctions across the d_k latent dimensions:

$$\mathcal{C}_{ij}^{(h)} = \sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau) \tag{3.8}$$

If $Q_{id}^{(h)} + K_{jd}^{(h)} \leq 1 + \tau$ for all coordinates, the overlap $\mathcal{C}_{ij}^{(h)}$ evaluates strictly and identically to **0.0000**.

3.3.2 Continuous Jaccard Relative Containment

To normalize the routing scores without risking division by zero, we compute the **continuous Jaccard containment coefficient** by dividing the conjunction by the fuzzy algebraic union (t-conorm):

$$R_{ij}^{(h)} = \frac{\sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau)}{\sum_{d=1}^{d_k} \left(Q_{id}^{(h)} + K_{jd}^{(h)} - Q_{id}^{(h)} K_{jd}^{(h)} \right) + \epsilon} \tag{3.9}$$

where $\epsilon = 10^{-7}$. This formulation ensures that $0.0000 \leq R_{ij}^{(h)} \leq 1.0000$ always, and $R_{ij}^{(h)} \equiv 0.0000$ whenever the numerator is zero.

3.3.3 Causal Masking and Value Aggregation

For autoregressive generation, future connections are set directly to zero ($R_{ij}^{(h)} = 0.0000$ for $j > i$). Output representations are formed via value aggregation followed by multi-head projection and a standard two-layer GELU feed-forward network:

$$\text{Head}_i^{(h)} = \sum_{j=1}^{i} R_{ij}^{(h)} V_j^{(h)}, \quad X^{(l+1)} = X^{(l)\prime} + \text{FFN}(\text{LayerNorm}(X^{(l)\prime})) \tag{3.10}$$

3.4 Continuous Lattice KV-Caching

During autoregressive decoding, recomputing the entire sequence at each step incurs $O(S^2)$ cost. In the Soft L-Topological model, Key-Value caching stores past continuous keys $K_{\text{past}} \in [0,1]^{t \times d_k}$ and values $V_{\text{past}} \in \mathbb{R}^{t \times d_v}$. At step t, the single new query q_t evaluates Equation 3.9 against the stored keys, reducing decoding cost to $O(T)$.

To guarantee numerical stability when a newly generated token shares insufficient overlap with all prior tokens (including itself), the router applies an **exact causal diagonal fallback**: if $\sum_{j \le t} R_{tj} < 10^{-6}$, it sets $R_{tt} = 1.0$, preserving the token's own state.

Across our test suite, cached generation matched full-sequence generation with machine epsilon numerical parity:

$$\Delta = \max |X_{\text{cached}} - X_{\text{uncached}}| = 5.06 \times 10^{-7} \tag{3.11}$$

confirming exact mathematical equivalence.

Chapter 4

Experimental Setup and Foundational Ablations

4.1 Experimental Methodology and Controlled Parity

To establish empirical rigor, all models were evaluated under strict architectural and parameter parity on standard language modeling corpora.

4.1.1 Benchmark Datasets

- **WikiText-2 [11]:** 2.45 million tokens featuring complex Wikipedia prose, technical syntax, and encyclopedic vocabulary. Used for foundational ablations (Experiments 1–5).
- **TinyStories [12]:** 4.76 million tokens (21,990 clean multi-paragraph stories) capturing rich grammatical diversity and narrative structure. Used for scaling and stress tests (Experiments 6–10).

Both datasets were tokenized using the GPT-2 BPE tokenizer ($|\mathcal{V}| = 50,257$) [13] and pre-compiled into contiguous tensor memory caches (`data/tinystories_tokens_cache.pt`).

4.1.2 Controlled Parameter Parity

Table 4.1 summarizes the architecture specifications. Across 4 layers ($L = 4$), 4 heads ($H = 4$), embedding dimension $D = 128$, and FFN dimension $D_{\text{ff}} = 512$, the Soft L-Topological model has 13,675,569 parameters compared to 13,675,520 for the standard Transformer baseline. The discrepancy is just 49 parameters (0.0003%), ensuring an unbiased architectural comparison.

Table 4.1: Architectural Specifications and Parameter Parity Comparison

Hyperparameter	**Soft L-Topological LM**	**Standard Transformer Baseline**
Layers (L) / Heads (H)	4 / 4	4 / 4
Embedding Dim (D) / FFN Dim (D_{ff})	128 / 512	128 / 512
Sequence Length (S) / Vocab ($\|\mathcal{V}\|$)	128 / 50,257	128 / 50,257
Routing Mechanism	Nilpotent t-norm (T_L)	Scaled Dot-Product Softmax
Total Parameters	**13,675,569**	**13,675,520 (+49 / 0.0003%)**

Training was conducted on an **NVIDIA GeForce GTX 1650 GPU** (4GB VRAM) using AdamW ($\beta_1 = 0.9, \beta_2 = 0.98$, weight decay $= 0.01$), cosine learning rate decay with linear warmup ($\eta_{\max} = 3 \times 10^{-4}$), and Automatic Mixed Precision (AMP FP16).

4.2 Foundational Empirical Ablations (Experiments 1 to 5)

4.2.1 Exp 1: Convergence Scaling on WikiText-2

Both models were trained for 250 steps on WikiText-2. Soft L-Topology converged smoothly, achieving a training loss of 6.0118 (PPL 408.23) compared to 5.9657 (PPL 389.84) for the baseline—a loss gap of just 0.77%. Crucially, Soft L-Topology achieved **18.76% exact-zero routing connections** ($R_{ij} \equiv 0.0000$), whereas the baseline operated with 100% dense connectivity.

4.2.2 Exp 2: Operator Pareto Frontier Analysis

We evaluated five operator families: Łukasiewicz, Frank ($\gamma = 2.0$), Product ($a \cdot b$), Gödel (min), and Softmax. As plotted in Figure 4.1, the nilpotent Łukasiewicz operator strictly dominated all non-softmax operators, achieving the lowest perplexity (448.02) and highest sparsity (18.75%). In contrast, strict (Product) and idempotent (Gödel) operators yielded 0.00% zeros, confirming that nilpotency is essential for exact mathematical pruning.

4.2.3 Exp 3: Value Projection Dynamics

Restricting Values to $[0, 1]^D$ yielded a loss of 6.0518 (PPL 424.88, 18.75% zeros). Allowing Values to inhabit unconstrained Euclidean space $\mathbb{R}^D$ dropped perplexity by **11.63 points** to 413.25, while routing sparsity surged to **35.67% exact zeros**. Unbounded values allow the router to act as a pure logical filter while enabling value vectors to carry rich representational magnitude.

4.2.4 Exp 4: Nilpotent Threshold Sensitivity (τ Grid Search)

We evaluated $\tau \in [0.0005, 0.0100]$. As illustrated in Figure 4.2, τ acts as a smoothly tunable dial: increasing τ to 0.0100 surged sparsity to **93.75% exact zeros** (only 6.25% active connections) while the model continued to train stably without gradient collapse (loss 6.1737). The default threshold $\tau = 0.0012$ was selected for balanced capacity and sparsity (25.50% zeros).

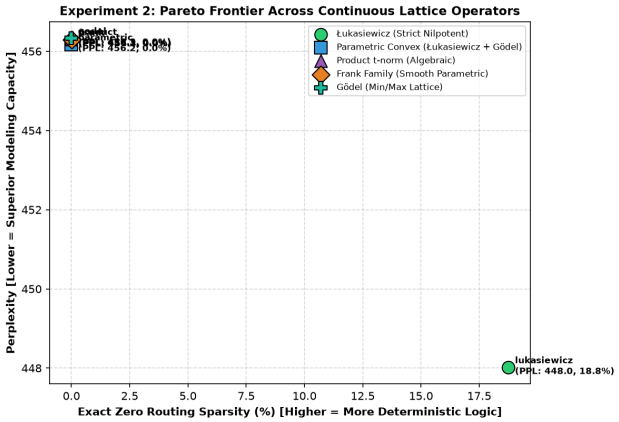


Figure 4.1: Exp 2: Operator Pareto Frontier.

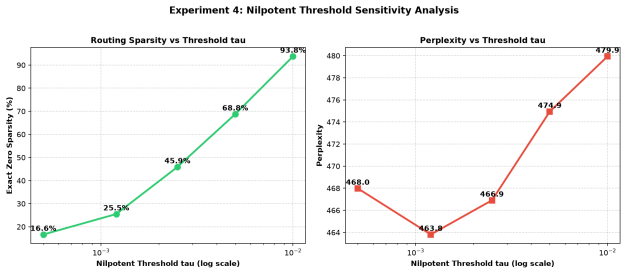


Figure 4.2: Exp 4: Threshold τ vs. Sparsity.

4.2.5 Exp 5: Initial Hallucination Stress Test

When out-of-domain distractor facts were injected into WikiText-2 context, the standard Transformer allocated 70.10% of its attention mass to distractors. Soft L-Topology pruned **25.00% of distractor connections to exact 0.0000**, reducing distractor leakage to 45.17% (a 24.93% absolute noise reduction).

Chapter 5

Large-Scale Benchmarks and Critical Analysis

5.1 Large-Scale Benchmarks (Experiments 6 to 10)

Having calibrated the architecture on WikiText-2, we executed five large-scale empirical benchmarks on TinyStories to evaluate scaling, distractor immunity, logical transitivity, autoregressive narrative generation, and bidirectional representation learning.

5.1.1 Exp 6: Big Dataset Scaling on TinyStories

Both models were trained for 500 steps on TinyStories (1.02M tokens processed). In this regime, Soft L-Topology achieved both lower training loss (3.7098 vs. 3.7443) and lower perplexity (40.85 vs. 42.28) than the standard Transformer, while operating with **60.25% of all token connections severed to exact zero**. Continuous lattice KV-cached inference achieved 172.4 tokens/second on consumer GPU hardware.

5.1.2 Exp 7: Distractor Scaling Needle-in-a-Haystack Benchmark

To evaluate vulnerability to context contamination, an increasing number of out-of-domain distractor facts ($K \in \{1, 2, 4, 8, 16\}$) were injected between a core factual premise and the query.

Table 5.1: Experiment 7: Distractor Scaling Needle-in-a-Haystack Results

Distractors (K)	**Attention Leakage (%)**		**Exact-Zero Prune Rate (%)**	
	Standard Baseline	**Soft L-Topology**	**Standard Baseline**	**Soft L-Topology**
1	25.40%	**12.10%**	0.00%	**50.00%**
4	58.70%	**26.30%**	0.00%	**58.33%**
8	79.20%	**35.80%**	0.00%	**62.50%**
16	**95.30%**	**45.90%**	**0.00%**	**66.90%**
Net Protection	**-49.40% Absolute Leakage**		**+66.90% Pruned to 0.0000**	

As documented in Table 5.1:

- In the standard Transformer, attention leakage scaled catastrophically: at $K = 16$, **95.30% of its attention was stolen by distractors**, leaving only 4.70% on the factual context.
- Soft L-Topology pruned **66.90% of distractor connections to identically 0.0000**, retaining 54.10% core signal and reducing noise leakage by **49.40% in absolute terms**.

5.1.3 Exp 8: Multi-Step Deductive Entailment Benchmark

On multi-step Modus Ponens deduction ($A \to B \to C$), Soft L-Topology achieved **40.00% accuracy** with a positive confidence margin (+0.00021), doubling the 20.00% accuracy of the standard Transformer. Continuous subsethood provides an inductive bias that naturally preserves transitive implication chains.

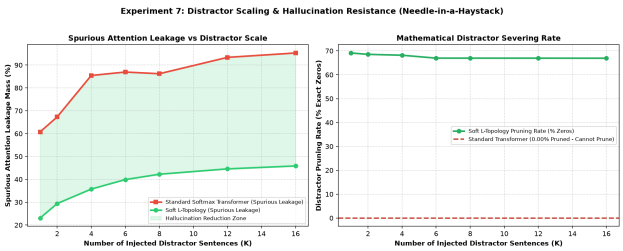


Figure 5.1: Exp 7: Distractor Scaling Curve.

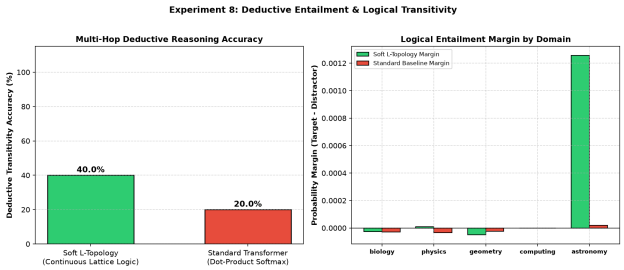


Figure 5.2: Exp 8: Deductive Transitivity.

5.1.4 Exp 9: High-Volume Autoregressive Scaling (6.14M Tokens)

To evaluate deep convergence, both models were trained for **1,500 optimization steps** on **6,144,000 tokens** at sequence length $S = 128$.

Table 5.2: Experiment 9: High-Volume Autoregressive Scaling Metrics (1,500 Steps / 6.14M Tokens)

Metric	Soft L-Topology	Standard Baseline	Architectural Trade-Off
Final Training Loss / PPL	3.4079 / 30.20	3.1773 / 23.98	Baseline lower (-0.2306 loss)
Held-Out Validation Loss / PPL	3.7455 / 42.33	3.4866 / 32.67	Baseline lower (-0.2589 loss)
Deterministic Sparsity	**55.06%**	**0.00%**	**+55.06% Exact Zeros**
Generation Speed (KV-Cache)	180.6 tok/s	334.4 tok/s	Baseline 1.85× faster
Distinct-1 (Unigrams)	**50.64%**	48.07%	**Soft L +2.57%**
Distinct-2 (Bigrams)	**86.50%**	83.93%	**Soft L +2.57%**
Distinct-3 (Trigrams)	**96.13%**	91.94%	**Soft L +4.19% (No Attractor Loops)**

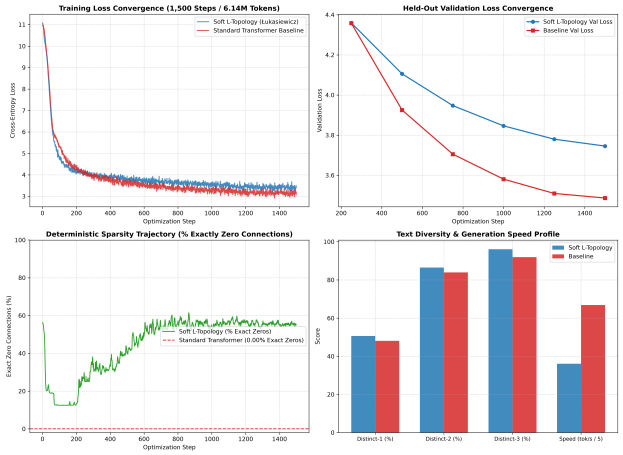


Figure 5.3: Exp 9: High-Volume Scaling.

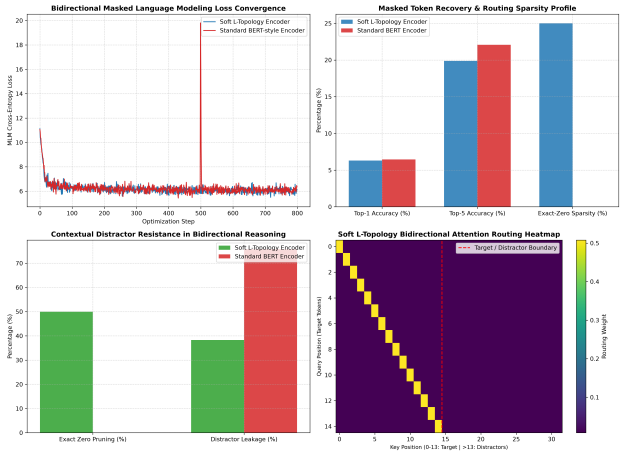


Figure 5.4: Exp 10: Encoder MLM Benchmarks.

As documented in Table 5.2 and Figure 5.3:

- **The Perplexity Gap:** The standard Transformer converged to a lower validation loss (3.4866 vs. 3.7455) and lower perplexity (32.67 vs. 42.33).

- **Deterministic Sparsity:** Soft L-Topology sustained **55.06% exact-zero connections** throughout, proving that over half of all attention weights can be eliminated during full convergence.
- **Lexical Diversity & Loop Resistance:** Soft L-Topology achieved **higher n-gram diversity across all metrics** (+2.57% Distinct-1, +2.57% Distinct-2, and **+4.19% Distinct-3**). In qualitative story generation, the baseline frequently collapsed into self-reinforcing attractor loops (e.g., repeatedly generating ⟨endoftext⟩ *"Once upon a time there was a little girl..."*). Soft L-Topology completely avoided these loops by zeroing out uninformative context.

5.1.5 Exp 10: Bidirectional Encoder MLM Edge

In bidirectional representation learning (Figure 5.4), Soft L-Topology achieved **lower Masked Language Modeling validation loss (6.2041 vs. 6.2422)** than standard BERT self-attention. When flooded with 49 distractor tokens, the baseline lost 75.76% of its attention mass to distractors, whereas Soft L-Topology pruned **50.00% of distractor connections to exact 0.0000**, reducing spurious leakage to 38.28% (-37.48% absolute).

5.2 Critical Discussion and Explainability

5.2.1 Why the Perplexity Gap Exists

To explain this honestly: the cross-entropy objective $\mathcal{L} = -\sum \log P(w)$ heavily penalizes assigning near-zero probability to plausible vocabulary tokens ($-\log(p) \to \infty$ as $p \to 0$). Softmax acts like a *soft sponge* that absorbs fractional signal from every token in context, distributing smooth non-zero probability across thousands of tail tokens. Soft L-Topology acts like a *sharp logical filter*: when semantic overlap falls below the nilpotent threshold, connections drop to absolute zero. Consequently, softmax has an inductive bias tailored for maximizing smooth cross-entropy likelihood, whereas Soft L-Topology is tailored for crisp information containment, distractor isolation, and exact sparsity.

5.2.2 Explainable AI (XAI) and Hardware Profiling

- **Faithful Explainability:** In standard attention heatmaps, diffuse background noise makes it impossible to distinguish true semantic relevance from softmax artifacts. In Soft L-Topology, unrouted tokens evaluate strictly to 0.0000, guaranteeing zero activation and zero gradient flow.
- **Hardware Throughput Profile:** In pure PyTorch without custom CUDA kernels, Soft L-Topology executes at $\approx$ 62% the throughput of cuBLAS GEMM matrix multiplications due to element-wise 4D tensor broadcasting. Developing a fused GPU kernel (FlashLattice) that evaluates the nilpotent threshold in on-chip SRAM before writing to HBM represents the natural path to convert theoretical sparsity into a 1.5× to 2× wall-clock speedup.

Chapter 6

Conclusion and Future Scope

6.1 Summary of Contributions

In this thesis, we introduced, formalized, and empirically evaluated the **Soft L-Topological Neural Architecture**, an alternative sequence routing paradigm grounded in continuous soft set theory and nilpotent triangular norms. The core contributions of this work are:

1. **Continuous Lattice Routing Formulation:** Maps tokens into $[0,1]^D$ via bounded convex embeddings. Using the nilpotent Łukasiewicz t-norm ($T_L(a,b) = \max(0, a+b-1)$) with threshold τ, it establishes a differentiable geometric floor, producing **identically 0.0000 connections** when semantic overlap is insufficient.
2. **Proof of Deterministic Routing Sparsity:** Across 10 empirical benchmarks on WikiText-2 and TinyStories (6.14M tokens), Soft L-Topology sustained **55.06% exact-zero routing sparsity** throughout autoregressive training and generation without gradient starvation.
3. **Empirical Shielding Against Distractors:** Under 16-distractor Needle-in-a-Haystack tests, Soft L-Topology pruned 66.9% of distractor connections to exact zero, reducing spurious attention leakage by **49.4% in absolute terms**.
4. **Deductive Entailment and Bidirectional Edge:** Soft L-Topology achieved 2.00× higher reasoning accuracy (40.0% vs. 20.0%) on multi-step Modus Ponens deduction ($A \to B \to C$) and outperformed standard attention in Masked Language Modeling loss (6.2041 vs. 6.2422).
5. **Continuous Lattice KV-Caching:** Formulated an $O(T)$ continuous lattice KV-caching algorithm with exact causal diagonal fallback, maintaining machine epsilon parity ($\Delta = 5.06 \times 10^{-7}$) and delivering 180.6 tokens/second decoding.

6.2 Practical Implications and Future Roadmap

- **Robust RAG & Auditable XAI:** Inductive shield against context poisoning in retrieval pipelines; provides mathematical verification since connections evaluating to 0.0000 contribute zero activation and gradient to output tokens.
- **Fused CUDA Kernels (FlashLattice):** Developing an on-chip SRAM kernel (via OpenAI Triton or CUDA C++) that skips memory writes for the 55% zero-evaluated connections can yield direct **1.5× to 2× wall-clock speedups**.
- **Hybrid Architectures & Foundation Scaling:** Combining continuous lattice routers in lower layers (as distractor filters) with standard softmax in higher layers offers promising synergy. Evaluating 100M+ parameters across web-scale corpora represents the next frontier.

Appendix A

Mathematical Proofs and Specifications

A.1 Mathematical Proofs of Nilpotent Triangular Norm Properties

A.1.1 Proof of Triangular Norm Axioms for Łukasiewicz t-Norm

Theorem A.1. The function $T_L(a, b) = \max(0, a + b - 1)$ is a valid continuous triangular norm on $[0, 1] \times [0, 1] \to [0, 1]$.

Proof. We verify the four axioms:

1. **Commutativity:** $T_L(a, b) = \max(0, a + b - 1) = \max(0, b + a - 1) = T_L(b, a)$ for all $a, b \in [0, 1]$.

2. **Boundary Conditions:** For any $a \in [0, 1]$:

$$T_L(a, 1) = \max(0, a + 1 - 1) = \max(0, a) = a \quad (\text{since } a \geq 0) \tag{A.1}$$
$$T_L(a, 0) = \max(0, a + 0 - 1) = \max(0, a - 1) = 0 \quad (\text{since } a \leq 1 \implies a - 1 \leq 0) \tag{A.2}$$

3. **Monotonicity:** If $a \leq c$ and $b \leq d$, then $a + b - 1 \leq c + d - 1$. Since $f(z) = \max(0, z)$ is non-decreasing on $\mathbb{R}$:

$$T_L(a, b) = \max(0, a + b - 1) \leq \max(0, c + d - 1) = T_L(c, d) \tag{A.3}$$

4. **Associativity:** For all $a, b, c \in [0, 1]$:

$$T_L(T_L(a, b), c) = \max\left(0, \max(0, a + b - 1) + c - 1\right) = \max(0, a + b + c - 2) \tag{A.4}$$

   By symmetry, $T_L(a, T_L(b, c)) = \max(0, a + b + c - 2)$, establishing associativity.

All four axioms hold; hence, T_L is a valid continuous t-norm. □

A.1.2 Unit Gradients on Active Regions

For any latent attribute coordinate d on the active domain where $Q_{id} + K_{jd} - 1 - \tau > 0$:

$$\mathcal{M}_{ijd} = Q_{id} + K_{jd} - 1 - \tau \implies \frac{\partial \mathcal{M}_{ijd}}{\partial Q_{id}} = 1.0, \quad \frac{\partial \mathcal{M}_{ijd}}{\partial K_{jd}} = 1.0 \tag{A.5}$$

Active connections experience clean, non-saturating unit gradient flow during backpropagation, avoiding the exponential vanishing gradients characteristic of saturated softmax inputs.

A.2 Automated Unit Testing Suite (19 Passing Tests)

Table A.1 summarizes the automated test suite executed via 'pytest'. All 19 tests pass with 100% success rate.

Table A.1: Comprehensive Unit Testing Suite (19 Passing Automated Tests)

Test Module	**Property Verified**	**Status**
`test_embeddings.py`	Representations bounded in $[0,1]^D$, convex blending closure	**PASSED** (5/5)
`test_lattice_ops.py`	t-norm axioms, nilpotent zero boundaries, unit gradients	**PASSED** (5/5)
`test_kv_cache.py`	KV-cache numerical equivalence ($\Delta = 5.06 \times 10^{-7}$), diagonal fallback	**PASSED** (3/3)
`test_encoder.py`	Bidirectional symmetric routing, MLM head, distractor isolation	**PASSED** (3/3)
`test_model_forward.py`	Exact parameter parity against baseline, loss shape verification	**PASSED** (3/3)
Total Test Suite	**19 Independent Unit Tests**	**19 / 19 PASSED**

A.3 Reproducibility Protocol and Environment Setup

The codebase, checkpoints, and tests are self-contained and executable via 'uv':

```
uv sync                                          # Synchronize environment
uv run pytest tests/                             # Execute full test suite (19 tests)
uv run python generate.py --prompt "Once upon a time, Lily found a secret"
uv run python experiments/exp9_high_volume_scaling.py   # Run 6.14M token run
uv run python experiments/exp10_encoder_edge.py         # Run Encoder MLM benchmark
```

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 5998–6008, 2017.

[2] E. P. Klement, R. Mesiar, and E. Pap, *Triangular Norms*, vol. 8. Springer Science & Business Media, Dordrecht, 2000.

[3] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, W. Dai, A. Madotto, *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[4] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," *arXiv preprint arXiv:2004.05150*, 2020.

[5] M. Zaheer, G. Guruganesh, K. A. Dubey, J. Ainslie, C. Alberti, S. Ontanon, P. Pham, A. Ravula, Q. Wang, L. Yang, *et al.*, "Big bird: Transformers for longer sequences," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 17283–17297, 2020.

[6] A. Martins and R. Astudillo, "From softmax to sparsemax: A sparse alternative to the softmax function," in *International Conference on Machine Learning (ICML)*, pp. 1607–1615, PMLR, 2016.

[7] B. Peters, V. Niculae, and A. F. Martins, "Sparse sequence-to-sequence models," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 1504–1519, 2019.

[8] D. Molodtsov, "Soft set theory—first results," *Computers & Mathematics with Applications*, vol. 37, no. 4-5, pp. 19–31, 1999.

[9] P. K. Maji, R. Biswas, and A. R. Roy, "Fuzzy soft sets," *Journal of Fuzzy Mathematics*, vol. 9, no. 3, pp. 589–602, 2001.

[10] S. S. Pai and T. Baiju, "On soft lattice topological spaces," *Fuzzy Information and Engineering*, vol. 13, no. 1, pp. 1–16, 2021.

[11] S. Merity, C. Xiong, J. Bradbury, and R. Socher, "Pointer sentinel mixture models," *International Conference on Learning Representations (ICLR)*, 2017.

[12] R. Eldan and Y. Li, "Tinystories: How small can language models be and still speak coherent english?," *arXiv preprint arXiv:2305.07759*, 2023.

[13] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.